6. VISCOSITY OF A LIQUID BY STOKE'S METHOD

AIM

To determine the co-efficient of viscosity of the given liquid by stoke's method

APPARATUS REQUIRED

A long cylindrical glass jar, highly viscous liquid, metre scale, spherical ball, stop clock, thread.

FORMULA

$$\eta = \frac{2r^2(\delta - \sigma)g}{9V} \text{ N s m}^{-2}$$

where η – Coefficient of viscosity of liquid (N s m^{-2})

$r \leftarrow$ radius of spherical ball (m)

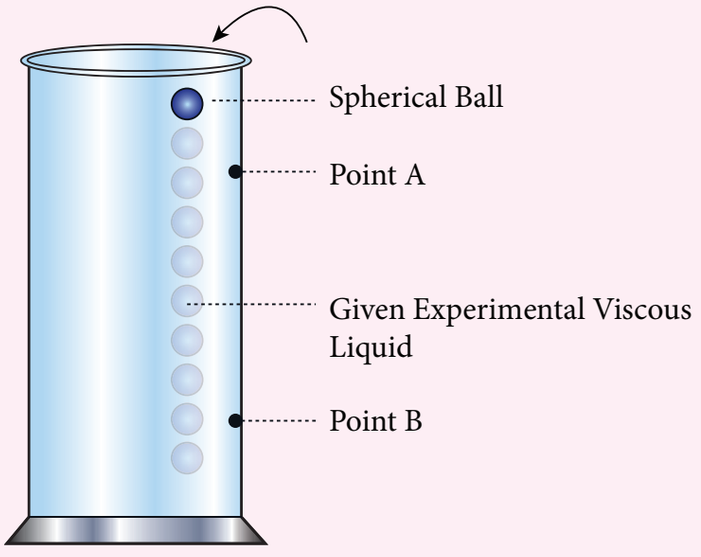
$\delta \leftarrow$ density of the steel sphere ($kg\ m^{-3}$)

$\sigma \leftarrow$ density of the liquid (kg m^{-3})

$g \leftarrow$ acceleration due to gravity (9.8 m s^{-2})

$V \leftarrow$ mean terminal velocity (m s^{-1})

DIAGRAM



EXPERIMENTAL SET UP OF MEASURING VISCOSITY BY STOKE'S METHOD

UNIT-12(XI-Physics_Vol-2) PRACTICAL FIRST YEAR.indd 295 02-01-2023 15:28:02